

Factsheet for the finite implication $E677 \models_{\text{fin}} E255$

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Abstract

Checked local facts for finite magmas satisfying

$$E677 : \quad x = y \diamond (x \diamond ((y \diamond x) \diamond y)), \quad E255 : \quad x = ((x \diamond x) \diamond x) \diamond x.$$

Fix x . Put $a = x \diamond x$, $q = a \diamond x$, and $p = x \backslash x$. The target is

$$E255 \text{ at } x \iff q \diamond x = x.$$

Everything below is proved here from E677 plus finiteness, or marked as external ETP input. No right cancellation, associativity, identity element, or quasigroup structure is used.

1 Setup and finite tools

A magma is a set with one binary operation $\diamond$. For $u \in M$, write

$$L_u(t) = u \diamond t, \quad R_u(t) = t \diamond u, \quad S(t) = t \diamond t.$$

If L_u is bijective, $u \backslash v$ denotes the unique w with $u \diamond w = v$. All indices on an L_x -cycle are read modulo its period.

Proposition 1 (Finite left division). *In every finite E677-magma, each L_a is bijective and*

$$a \backslash b = b \diamond ((a \diamond b) \diamond a).$$

Also

$$a \backslash (a \backslash b) = (a \backslash b) \diamond (b \diamond a), \quad b \backslash (a \backslash b) = (a \diamond b) \diamond a.$$

Proof. Proof certificate. E677 gives

$$b = a \diamond (b \diamond ((a \diamond b) \diamond a)),$$

so L_a is surjective, hence bijective by finiteness. Substitute $a \backslash b$ for b , and use uniqueness of left division. $\square$

Lemma 2 (Two global identities). *For all y, z ,*

$$z = (y \diamond z) \diamond ((y \diamond (y \diamond z)) \diamond y) \quad (\text{T})$$

and

$$(y \diamond (y \diamond z)) \diamond y = z \diamond (((y \diamond z) \diamond z) \diamond (y \diamond z)). \quad (\text{K})$$

Proof. Proof certificate. For (T), substitute $y \diamond z$ for the law-variable x in E677, then left-cancel y . For (K), compare (T) with E677 using parameter $y \diamond z$, then left-cancel $y \diamond z$. $\square$

Remark 3 (External ETP guardrails). The unrestricted implication $\text{E677} \models \text{E255}$ is false in the free E677-magma, so any proof must use finiteness. Linear models satisfying E677 satisfy E255; affine-fiber extensions over them are ruled out by ETP structure theory. Finite E677-magmas need not be right-cancellative [3].

2 Local equivalences

Fix a finite E677-magma M and $x \in M$. Put

$$a = x \diamond x, \quad q = a \diamond x = (x \diamond x) \diamond x, \quad p = x \backslash x.$$

Proposition 4 (Unique right fixed-point witness). *If $u \diamond x = x$, then $u = q$. Hence*

$$\text{E255 at } x \iff \exists u (u \diamond x = x) \iff q \diamond x = x.$$

Proof. Proof certificate. If $u \diamond x = x$, then (T) gives $x = x \diamond (x \diamond u)$. The specialization $(x, y) = (x, x)$ in E677 gives $x = x \diamond (x \diamond q)$. Two left cancellations give $u = q$. $\square$

Proposition 5 (Equivalent local targets). *The following are equivalent:*

$$\text{E255 at } x, \quad q \diamond x = x, \quad q \backslash x = x, \quad ((x \diamond x) \diamond q) \diamond (x \diamond x) = x, \quad (q \diamond x) \diamond q = p.$$

The fourth identity is the identity sometimes called Lemma L.

Proof. Proof certificate. The first two are Proposition 4. (T), with $(y, z) = (x \diamond x, x)$, gives

$$x = q \diamond (((x \diamond x) \diamond q) \diamond (x \diamond x)).$$

Thus $q \backslash x = ((x \diamond x) \diamond q) \diamond (x \diamond x)$. Finally

$$q \backslash x = x \diamond ((q \diamond x) \diamond q),$$

and left cancellation by x compares this with $p = x \backslash x$. $\square$

Lemma 6 (Fixed maps). *Define*

$$H_x(t) = ((x \diamond t) \diamond x), \quad F_x(t) = x \diamond (t \diamond x).$$

Then

$$x \setminus t = t \diamond H_x(t), \quad F_x(L_x(t)) = L_x(H_x(t)).$$

Thus F_x and H_x have the same finite cycle structure. Moreover E255 at x is equivalent to either map having a fixed point. If fixed points exist, they are unique:

$$\text{Fix}(F_x) = \{x \setminus q\}, \quad \text{Fix}(H_x) = \{x \setminus (x \setminus q)\}.$$

Under $q \diamond x \neq x$, both maps are fixed-point-free.

Proof. Proof certificate. E677 with parameter x gives

$$t = x \diamond (t \diamond ((x \diamond t) \diamond x)) = x \diamond (t \diamond H_x(t)),$$

hence $x \setminus t = t \diamond H_x(t)$. Also

$$F_x(L_x(t)) = x \diamond ((x \diamond t) \diamond x) = x \diamond H_x(t) = L_x(H_x(t)),$$

so $F_x = L_x H_x L_x^{-1}$, and F_x, H_x are conjugate.

Suppose $F_x(t) = t$. Comparing

$$x \diamond (t \diamond x) = t = x \diamond (t \diamond H_x(t))$$

and left-cancelling first x and then t , we get $H_x(t) = x$. Thus $(x \diamond t) \diamond x = x$, so Proposition 4 gives $x \diamond t = q$; hence $t = x \setminus q$. Conversely, if $q \diamond x = x$ and $t = x \setminus q$, then $H_x(t) = (x \diamond t) \diamond x = q \diamond x = x$, and the displayed edge formula gives

$$t = x \diamond (t \diamond x) = F_x(t).$$

Thus F_x has a fixed point exactly when E255 at x holds, and then its only fixed point is $x \setminus q$. The statement for H_x follows by conjugacy: $H_x(s) = s$ iff $F_x(L_x(s)) = L_x(s)$. Hence, when fixed points exist,

$$L_x(s) = x \setminus q, \quad s = x \setminus (x \setminus q).$$

If $q \diamond x \neq x$, Proposition 4 says that no right fixed point over x exists, so neither F_x nor H_x has a fixed point. $\square$

Lemma 7 (Division return trap). *For all y, z , put*

$$u = y \diamond z, \quad v = u \diamond y.$$

Then

$$v = z \setminus (y \setminus z), \quad y \setminus z = z \diamond v. \tag{D}$$

Consequently, if also $z \diamond v = u$, then

$$z = y \diamond u.$$

Proof. Proof certificate. The formula for $y \setminus z$ is just the left-division form of E677:

$$y \setminus z = z \diamond ((y \diamond z) \diamond y) = z \diamond v.$$

Left-dividing this equality by z gives

$$v = z \setminus (y \setminus z).$$

If $z \diamond v = u$, then the displayed identity gives $y \setminus z = u$. Left-multiply by y to obtain $z = y \diamond u$. $\square$

3 Orbit facts

Define

$$c_0 = x, \quad c_{k+1} = x \diamond c_k,$$

and let d be the minimal period of the L_x -orbit of x . Put

$$d_k = c_k \diamond x.$$

Proposition 8 (Basic orbit package). *For all k ,*

$$\begin{aligned} c_{k-1} &= c_k \diamond d_{k+1}, & d_1 &= c_{d-2}, \\ x \setminus c_k &= c_{k-1}, & c_k \setminus c_{k-1} &= d_{k+1}. \end{aligned}$$

In particular

$$p = c_{d-1}, \quad q = x \setminus p = d_1 = c_{d-2} = a \diamond x, \quad p \diamond a = q, \quad p \setminus q = a.$$

If $d_j = x$, then $j \equiv d - 2 \pmod{d}$.

Proof. Proof certificate. Applying E677 with parameter x and law-variable c_k gives

$$c_k = x \diamond (c_k \diamond ((x \diamond c_k) \diamond x)) = x \diamond (c_k \diamond d_{k+1}).$$

Since also $c_k = x \diamond c_{k-1}$, left cancellation by x gives

$$c_{k-1} = c_k \diamond d_{k+1}.$$

The equality $x \diamond q = p$ follows from the specialization

$$x = x \diamond (x \diamond q)$$

of E677 at $(x, y) = (x, x)$, because $p = x \setminus x$. Therefore $q = x \setminus p = c_{d-2}$, while $q = a \diamond x = d_1$ by definition. The division statements follow from $x \diamond c_{k-1} = c_k$ and $c_k \diamond d_{k+1} = c_{k-1}$. In particular $p = c_{d-1}$, $p \diamond a = q$ is the recurrence at $k = d - 1$, and $p \setminus q = a$.

Finally suppose $d_j = x$. By (T) with $(y, z) = (c_j, x)$,

$$x = x \diamond ((c_j \diamond x) \diamond c_j) = x \diamond (x \diamond c_j).$$

Comparing with $x = x \diamond (x \diamond q)$ and left-cancelling x twice gives $c_j = q$. Since $q = c_{d-2}$ on the L_x -cycle, $j \equiv d - 2 \pmod{d}$. $\square$

Corollary 9 (Small periods). *If $d = 1$, then E255 at x holds. The cases $d = 2$ and $d = 3$ are impossible. Thus a bad point has $d \geq 4$.*

Proof. Proof certificate. $d = 1$ means $x \diamond x = x$, so $q = x$ and $q \diamond x = x$.

If $d = 2$, then $q = c_{d-2} = c_0 = x$, hence $d_1 = q \diamond x = x$, contradicting the forced-index statement in Proposition 8, since $1 \not\equiv 0 \pmod{2}$.

Assume $d = 3$. Then

$$c_0 = x, \quad c_1 = a = q, \quad c_2 = p,$$

and

$$a \diamond x = a, \quad x \diamond a = p, \quad x \diamond p = x, \quad p \diamond a = a.$$

The orbit recurrence at $k = 1$ gives

$$x = a \diamond (p \diamond x). \tag{D3.1}$$

By (K) with $(y, z) = (x, a)$,

$$a = (x \diamond (x \diamond a)) \diamond x = a \diamond (((x \diamond a) \diamond a) \diamond (x \diamond a)) = a \diamond (a \diamond p).$$

Since also $a \diamond x = a$, left cancellation by a gives

$$a \diamond p = x. \tag{D3.2}$$

Comparing (D3.1) with (D3.2) and left-cancelling a gives

$$p \diamond x = p. \tag{D3.3}$$

Now (K) with $(y, z) = (p, a)$ gives

$$x = (p \diamond (p \diamond a)) \diamond p = a \diamond (((p \diamond a) \diamond a) \diamond (p \diamond a)) = a \diamond ((a \diamond a) \diamond a).$$

Since $a \diamond p = x$, left cancellation by a gives

$$(a \diamond a) \diamond a = p. \tag{D3.4}$$

Finally, (K) with $(y, z) = (a, x)$ gives

$$(a \diamond a) \diamond a = x \diamond (a \diamond a).$$

Using (D3.4) and $x \diamond a = p$, left cancellation by x gives $a \diamond a = a$. Then $(a \diamond a) \diamond a = a$, contradicting (D3.4) and the distinctness of $a = c_1$ and $p = c_2$ on a period-three orbit. $\square$

Lemma 10 (Bad orbit-edge exclusions). *Assume E255 at x fails. Then for every k ,*

$$d_k = c_k \diamond x \notin \{x, c_{k-1}\},$$

or equivalently

$$c_k \diamond x \neq x, \quad x \diamond (c_k \diamond x) \neq c_k.$$

Proof. Proof certificate. The first exclusion is Proposition 4: under failure, no element u satisfies $u \diamond x = x$. If $d_k = c_{k-1}$, then

$$H_x(c_{k-1}) = ((x \diamond c_{k-1}) \diamond x) = c_k \diamond x = d_k = c_{k-1},$$

contradicting the fixed-point-free conclusion of Lemma 6. Since $x \diamond c_{k-1} = c_k$, left cancellation by x makes $d_k \neq c_{k-1}$ equivalent to $x \diamond d_k \neq c_k$. $\square$

Lemma 11 (Uniform orbit-edge identities). *For all k ,*

$$d_{k+2} = c_k \diamond ((c_{k+1} \diamond c_k) \diamond c_{k+1}), \quad (\text{U1})$$

$$x = d_k \diamond ((c_k \diamond d_k) \diamond c_k), \quad (\text{U2})$$

$$(c_k \diamond d_k) \diamond c_k = x \diamond ((d_k \diamond x) \diamond d_k). \quad (\text{U3})$$

Therefore

$$d_k \backslash x = (c_k \diamond d_k) \diamond c_k = x \diamond ((d_k \diamond x) \diamond d_k).$$

At a bad point,

$$a \backslash x = d_2, \quad q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q), \quad q \backslash a = (p \diamond q) \diamond p.$$

Proof. Proof certificate. (U1) is (K) with $(y, z) = (x, c_k)$. (U2) is (T) with $(y, z) = (c_k, x)$. (U3) is (K) with $(y, z) = (c_k, x)$. The quotient forms read (U2) as left division and use (U3). For the displayed specializations, $a \backslash x = x \diamond ((a \diamond x) \diamond a) = x \diamond (q \diamond a)$, and (U1) at $k = 0$ says $d_2 = x \diamond (q \diamond a)$. Since $d_1 = q$, the general quotient formula at $k = 1$ gives

$$q \backslash x = (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q).$$

Finally $p \diamond a = q$, and (T) with $(y, z) = (p, a)$ gives

$$a = q \diamond ((p \diamond q) \diamond p),$$

so $q \backslash a = (p \diamond q) \diamond p$. $\square$

Proposition 12 (Period-four gate). *Assume $d = 4$. Write*

$$a = c_1, \quad q = c_2, \quad p = c_3, \quad r = q \diamond x, \quad s = p \diamond x.$$

Then

$$x \diamond a = q, \quad a \diamond x = q, \quad a \diamond r = x, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q,$$

and $r \notin \{x, a, q\}$. Thus $r = p$ or $r \notin \{x, a, q, p\}$. Further,

$$q \diamond s = a, \quad (p \diamond q) \diamond p = s.$$

If $r = p$, then

$$q \diamond a = q, \quad s = (q \diamond q) \diamond q = a \diamond (q \diamond q), \quad s \notin \{x, a, q\},$$

$$H_x(a) = p, \quad H_x(p) = a.$$

If E255 at x fails and $d = 4$, then either

$$q \diamond x \notin \{x, a, q, p\},$$

or

$$q \diamond x = p \quad \text{and} \quad p \diamond x \notin \{x, a, q, p\}.$$

In particular, at least one of $q \diamond x$ and $p \diamond x$ lies outside $\{x, a, q, p\}$.

Proof. Proof certificate. The L_x -orbit identities give

$$x \diamond a = q, \quad x \diamond q = p, \quad x \diamond p = x.$$

Since $q = d_1 = a \diamond x$, we also have $a \diamond x = q$. The recurrence $c_{k-1} = c_k \diamond d_{k+1}$, for $k = 1, 2, 3$, gives

$$a \diamond r = x, \quad q \diamond s = a, \quad p \diamond a = q.$$

We next exclude $r = x, a, q$. If $r = x$, then $a \diamond r = a \diamond x = q$, but $a \diamond r = x$, so $q = x$, impossible on a period-four orbit.

If $r = a$, then (U3) at $k = 0$ gives

$$a = r = x \diamond (q \diamond a).$$

Since $x \diamond x = a$, left cancellation by x gives $q \diamond a = x$. Now (K) with $(y, z) = (q, x)$ gives

$$(q \diamond r) \diamond q = x \diamond ((r \diamond x) \diamond r).$$

With $r = a$, this becomes

$$(q \diamond a) \diamond q = x \diamond ((a \diamond x) \diamond a),$$

hence $p = a$, because the left side is $x \diamond q = p$ and the right side is $x \diamond (q \diamond a) = x \diamond x = a$. This is impossible.

If $r = q$, then $a \diamond q = x$. By (U3) at $k = 1$,

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q),$$

so $q = x \diamond (q \diamond q)$. Comparing with $q = x \diamond a$ gives $q \diamond q = a$. Applying (K) with $(y, z) = (q, x)$ and using $r = q$ gives

$$(q \diamond q) \diamond q = x \diamond (q \diamond q).$$

The left side is $a \diamond q = x$, while the right side is $x \diamond a = q$, another contradiction. Thus $r \notin \{x, a, q\}$, so either $r = p$ or r is outside the principal orbit $\{x, a, q, p\}$.

The identity $q \diamond s = a$ was obtained above, and

$$(p \diamond q) \diamond p = s$$

is the specialization $q \setminus a = (p \diamond q) \diamond p$ from Lemma 11, because $q \diamond s = a$.

Assume now that $r = p$. From (U3) at $k = 0$,

$$p = r = x \diamond (q \diamond a).$$

Since $x \diamond q = p$, left cancellation by x gives

$$q \diamond a = q. \quad (\text{P4.1})$$

Then (U1) at $k = 1$ gives

$$s = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q). \quad (\text{P4.2})$$

Also (K) with $(y, z) = (q, a)$, together with (P4.1), gives

$$(q \diamond q) \diamond q = a \diamond ((q \diamond a) \diamond q) = a \diamond (q \diamond q) = s. \quad (\text{P4.3})$$

The exclusions for s are immediate. If $s = x$, then $p \diamond x = x$, so Proposition 4 gives $p = q$. If $s = a$, then

$$a = q \diamond s = q \diamond a = q$$

by (P4.1). If $s = q$, then $p \diamond x = q = p \diamond a$, and left cancellation by p gives $x = a$. Hence $s \notin \{x, a, q\}$.

Finally,

$$H_x(a) = ((x \diamond a) \diamond x) = q \diamond x = r = p, \quad H_x(p) = ((x \diamond p) \diamond x) = x \diamond x = a.$$

If E255 at x fails, then $r = q \diamond x \neq x$. The preceding exclusions give either $r \notin \{x, a, q, p\}$, in which case we are done, or $r = p$. In the latter case the exclusions above already give $s = p \diamond x \notin \{x, a, q\}$. It remains only to exclude $s = p$. Suppose $s = p$, and put $t = q \diamond q$. Then (P4.2) and (P4.3) give

$$a \diamond t = p, \quad t \diamond q = p.$$

Also $p \diamond a = q$. Lemma 7, applied with

$$y = a, \quad z = t, \quad u = p, \quad v = q,$$

therefore gives

$$t = a \diamond p = x.$$

But then $p = a \diamond t = a \diamond x = q$, contradicting the distinctness of p and q on the period-four orbit. Hence $s \neq p$, and so $s \notin \{x, a, q, p\}$. $\square$

Proposition 13 (Period-five and period-six first data). *If $d = 5$, write*

$$a = c_1, \quad b = c_2, \quad q = c_3, \quad p = c_4, \quad g = q \diamond x.$$

Then

$$\begin{aligned} a \diamond x &= q, & x \diamond q &= p, & x \diamond p &= x, & p \diamond a &= q, \\ g &= a \diamond ((b \diamond a) \diamond b), & q \diamond ((a \diamond q) \diamond a) &= x, & (a \diamond q) \diamond a &= x \diamond (g \diamond q), \end{aligned}$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \quad g \notin \{x, b\}, \quad p \diamond x \notin \{x, q\}.$$

If $d = 6$, write

$$a = c_1, \quad b = c_2, \quad c = c_3, \quad q = c_4, \quad p = c_5, \quad h = c \diamond x, \quad g = q \diamond x.$$

Then

$$\begin{aligned} a \diamond x &= q, & x \diamond q &= p, & x \diamond p &= x, & p \diamond a &= q, \\ h &= a \diamond ((b \diamond a) \diamond b), & g &= b \diamond ((c \diamond b) \diamond c), \\ q \diamond ((a \diamond q) \diamond a) &= x, & (a \diamond q) \diamond a &= x \diamond (g \diamond q), \end{aligned}$$

and at a bad point

$$b \diamond x \notin \{x, a\}, \quad h \notin \{x, b\}, \quad g \notin \{x, c\}, \quad p \diamond x \notin \{x, q\}.$$

Proof. Proof certificate. Suppose first that $d = 5$. Then $q = c_{d-2} = c_3$ and $p = c_{d-1} = c_4$, so

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x.$$

The recurrence at $k = 4$ gives $p \diamond a = q$. By (U1) at $k = 1$,

$$g = d_3 = a \diamond ((b \diamond a) \diamond b).$$

By (U2) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x,$$

and (U3) at $k = 1$ gives

$$(a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point, $g = x$ would be $q \diamond x = x$, contrary to the equivalence in Proposition 4; the remaining exclusions are exactly Lemma 10 at $k = 2, 3, 4$.

The $d = 6$ case is the same calculation with $q = c_4$ and $p = c_5$. Thus

$$a \diamond x = q, \quad x \diamond q = p, \quad x \diamond p = x, \quad p \diamond a = q.$$

By (U1) at $k = 1$ and $k = 2$,

$$h = d_3 = a \diamond ((b \diamond a) \diamond b), \quad g = d_4 = b \diamond ((c \diamond b) \diamond c).$$

By (U2) and (U3) at $k = 1$,

$$q \diamond ((a \diamond q) \diamond a) = x, \quad (a \diamond q) \diamond a = x \diamond ((q \diamond x) \diamond q) = x \diamond (g \diamond q).$$

At a bad point the displayed exclusions are Lemma 10 at $k = 2, 3, 4, 5$. □

4 Collision and right-tail facts

Proposition 14 (Collision lift). *If $d_i = d_j = e$, then*

$$(c_i \diamond e) \diamond c_i = (c_j \diamond e) \diamond c_j.$$

If also $i \not\equiv j \pmod{d}$, then

$$c_i \diamond e \neq c_j \diamond e.$$

Proof. Proof certificate. (T) gives $x = d_k \diamond ((c_k \diamond d_k) \diamond c_k)$. Equal d_i, d_j allow left cancellation by e . If also $c_i \diamond e = c_j \diamond e = s$, then left-cancel s to get $c_i = c_j$. $\square$

Proposition 15 (Right-collision rectangles). *If $y \diamond x = r$, then*

$$(y \diamond r) \diamond y = r \setminus x. \tag{R}$$

Consequently, if $y \diamond x = z \diamond x = r$ and $y \diamond r = z \diamond r$, then $y = z$.

Proof. Proof certificate. (T) with $(y, z) = (y, x)$ gives $x = r \diamond ((y \diamond r) \diamond y)$. This is (R). The last claim is left cancellation after applying (R) to y and z . $\square$

Lemma 16 (Right-tail first repetition). *Assume $q \diamond x \neq x$. Define*

$$\rho_0 = x, \quad \rho_{n+1} = \rho_n \diamond x.$$

Then $\rho_n \neq x$ for $n \geq 1$. Let $1 \leq i < j$ be the first repetition in the tail, with j minimal, and put

$$e = \rho_i = \rho_j, \quad y = \rho_{i-1}, \quad z = \rho_{j-1}.$$

Then

$$\begin{aligned} y &\neq z, & y \diamond x &= z \diamond x = e, \\ (y \diamond e) \diamond y &= (z \diamond e) \diamond z = e \setminus x, & y \diamond e &\neq z \diamond e. \end{aligned}$$

Proof. Proof certificate. If some $\rho_n = x$, then $\rho_{n-1} \diamond x = x$, forcing $\rho_{n-1} = q$, contradiction. Finiteness gives a first repetition. Minimality gives $y \neq z$. Proposition 15 gives the rectangle and splitter. $\square$

Proposition 17 (Left-translation labels). *Write $a \diamond b = \sigma(a)(b)$. Then $\sigma(a) \in \text{Sym}(M)$ and the label map $a \mapsto \sigma(a)$ is injective. If $y \diamond x = z \diamond x = r$ with $y \neq z$, then*

$$\begin{aligned} \sigma(y)(x) &= \sigma(z)(x) = r, & \sigma(y)(r) &\neq \sigma(z)(r), \\ \sigma(\sigma(y)(r))(y) &= r \setminus x = \sigma(\sigma(z)(r))(z). \end{aligned}$$

Proof. Proof certificate. Left translations are permutations by Proposition 1. Suppose $L_y = L_z$. Choose any $x \in M$ and put

$$e = y \diamond x = z \diamond x.$$

The two E677 specializations give

$$x = y \diamond (x \diamond (e \diamond y)), \quad x = z \diamond (x \diamond (e \diamond z)).$$

Since $L_y = L_z$, the second equality may be written as

$$x = y \diamond (x \diamond (e \diamond z)).$$

Comparing with the first equality and left-cancelling successively by y , then by x , then by e , gives $y = z$. Thus $a \mapsto \sigma(a)$ is injective.

If $y \diamond x = z \diamond x = r$ and $y \neq z$, Proposition 15 gives

$$(y \diamond r) \diamond y = (z \diamond r) \diamond z = r \setminus x.$$

It also says that $y \diamond r = z \diamond r$ would force $y = z$, so $\sigma(y)(r) \neq \sigma(z)(r)$. Rewriting these products in the notation $a \diamond b = \sigma(a)(b)$ gives the displayed formulas. $\square$

5 Stress-tested candidate structures

A candidate structure in this section is only a partial local diagram around a bad point x . It is not asserted to extend to a finite magma unless explicitly stated. The word “survives” means that the currently proved tests below do not yet eliminate the diagram, and the displayed equations are the obligations that remain.

Proposition 18 (Universal stress tests). *Assume E255 at x fails. Then every candidate around x must pass the following tests.*

1. R_x is not surjective and not injective. In particular, no finite right-cancellative, quasi-group, or loop candidate can be bad at x .
2. No commutative finite candidate can be bad at x .
3. The L_x -period of x is $d \geq 4$. In particular, $x \diamond x \neq x$, and the periods $d = 2$ and $d = 3$ are unavailable.
4. For every orbit edge,

$$d_k = c_k \diamond x \notin \{x, c_{k-1}\}.$$

5. Every first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$ produces a splitter rectangle: if $y \diamond x = z \diamond x = e$ with $y \neq z$, then

$$(y \diamond e) \diamond y = (z \diamond e) \diamond z = e \setminus x, \quad y \diamond e \neq z \diamond e.$$

6. Every proposed short return must pass the division return trap of Lemma 7: if $u = y \diamond z$, $v = u \diamond y$, and $z \diamond v = u$, then $z = y \diamond u$.

Proof. Proof certificate. At a bad point, Proposition 4 gives no solution to $u \diamond x = x$, so $x \notin \text{im } R_x$. Since the magma is finite, an injective R_x would be surjective. This proves the first item. If the operation is commutative, then $R_x = L_x$, but L_x is bijective by Proposition 1; this proves the second item. The period statement is Corollary 9. The orbit-edge exclusion is Lemma 10. The splitter rectangle is Lemma 16, or Proposition 15 for any chosen collision fiber. The last item is Lemma 7. $\square$

Proposition 19 (Period-five internal-tail stress tests). *Assume $d = 5$ and E255 at x fails. Write*

$$a = c_1, \quad b = c_2, \quad q = c_3, \quad p = c_4,$$

and put

$$e = b \diamond x, \quad g = q \diamond x, \quad s = p \diamond x.$$

Then

$$e \notin \{x, a\}, \quad g \notin \{x, b\}, \quad s \notin \{x, q\},$$

and

$$b \diamond g = a, \quad q \diamond s = b, \quad p \diamond a = q. \quad (\text{P5.0})$$

If the right tail stays in the principal orbit through g , then the only possibilities are $g \in \{a, q, p\}$, and they have the following stress data.

1. If $g = q$, then

$$b \diamond q = a, \quad (a \diamond q) \diamond a = (q \diamond q) \diamond q = q \setminus x, \quad a \diamond q \neq q \diamond q.$$

2. If $g = a$, then

$$b \diamond a = a, \quad (q \diamond a) \diamond q = b \diamond x = a \setminus x, \quad q \diamond a \neq b.$$

3. If $g = p$, then

$$b \diamond p = a, \quad q \diamond s = b, \quad s \in M \setminus \{x, q\}.$$

In the internal subcases for s , the following extra tests apply:

$$\begin{aligned} s = a &\implies \begin{aligned} q \diamond a &= b, \\ q \diamond p &= b \diamond x = a \setminus x, \end{aligned} \\ s = p &\implies \begin{aligned} q \diamond p &= b, \\ b \diamond q &= (p \diamond p) \diamond p = p \setminus x, \quad q \diamond p \neq p \diamond p, \end{aligned} \\ s = b &\implies q \diamond b = b. \end{aligned}$$

If moreover $s = b$ and the next value $e = b \diamond x$ is still principal-orbit-internal, then $e \in \{b, q, p\}$, giving respectively the collision fibers

$$\{p, b\} \rightarrow b, \quad \{a, b\} \rightarrow q, \quad \{q, b\} \rightarrow p.$$

Proof. Proof certificate. The exclusions are Proposition 13. The recurrence $c_{k-1} = c_k \diamond d_{k+1}$, with $d_3 = g$ and $d_4 = s$, gives

$$b \diamond g = a, \quad q \diamond s = b,$$

and the recurrence at $k = 4$ gives $p \diamond a = q$.

If $g = q$, then $b \diamond q = a$ follows from $b \diamond g = a$. Also $a \diamond x = q = q \diamond x$, so Proposition 15, applied to the fiber $\{a, q\} \rightarrow q$, gives the displayed splitter rectangle.

If $g = a$, then $b \diamond a = a$. Since $x \diamond x = a = q \diamond x$, Proposition 15 gives

$$(x \diamond a) \diamond x = (q \diamond a) \diamond q = a \setminus x, \quad x \diamond a \neq q \diamond a.$$

Because $x \diamond a = b$, this is the displayed statement.

If $g = p$, then $b \diamond p = a$, and the relation $q \diamond s = b$ is already in (P5.0). The exclusion $s \notin \{x, q\}$ is the bad orbit-edge exclusion at $k = 4$. If $s = a$, then $q \diamond a = b$, and the fiber $\{x, p\} \rightarrow a$ gives

$$(x \diamond a) \diamond x = (p \diamond a) \diamond p = a \setminus x.$$

Using $x \diamond a = b$ and $p \diamond a = q$, this becomes $q \diamond p = b \diamond x = a \setminus x$. If $s = p$, then $q \diamond p = b$, and the fiber $\{q, p\} \rightarrow p$ gives

$$(q \diamond p) \diamond q = (p \diamond p) \diamond p = p \setminus x, \quad q \diamond p \neq p \diamond p,$$

which is the displayed statement. If $s = b$, then $q \diamond b = b$. Finally, when $s = b$, the next right-tail value is $e = b \diamond x$. If it is principal-orbit-internal, then $e \notin \{x, a\}$, so $e \in \{b, q, p\}$; comparing with the earlier right-tail values $a \diamond x = q$, $q \diamond x = p$, and $p \diamond x = b$ gives the three listed fibers. $\square$

Remark 20 (Candidate pruning ledger). The current hand-pruned candidate zoo is as follows. A ruled-out line is eliminated by the cited stress tests; a surviving line is only a local blueprint for the next pruning pass.

- C1. *Finite right-cancellative, quasigroup, or loop candidate: ruled out.* At a bad point, R_x misses x , hence cannot be injective in a finite magma.
- C2. *Commutative finite candidate: ruled out.* Commutativity gives $R_x = L_x$, contradicting the missing value $x \notin \text{im } R_x$.
- C3. *Idempotent bad point: ruled out.* If $x \diamond x = x$, then $d = 1$, so $q = x$ and $q \diamond x = x$.
- C4. *Small-period candidate $d = 2$ or $d = 3$: ruled out.* These are exactly the impossible cases in Corollary 9.
- C5. *Period-four candidate on $O_4 = \{x, a, q, p\}$ only: ruled out.* Proposition 12 forces $q \diamond x$ or $p \diamond x$ outside O_4 .
- C6. *Period-four delayed loop $q \diamond x = p$, $p \diamond x = p$: ruled out.* This is the short-return branch killed by Lemma 7 inside the proof of Proposition 12.
- C7. *Period-four early external exit: survives.* Here

$$q \diamond x = r \notin O_4, \quad a \diamond r = x,$$

with $x \diamond q = p$, $x \diamond p = x$, and $p \diamond a = q$. The next tests are the splitter rectangle at the first repetition of the right tail and the fixed-point gate for the external element r .

- C8. *Period-four delayed external exit: survives.* Here

$$q \diamond x = p, \quad p \diamond x = s \notin O_4,$$

and Proposition 12 gives

$$q \diamond a = q, \quad q \diamond s = a, \quad p \diamond a = q, \quad s = (q \diamond q) \diamond q = a \diamond (q \diamond q).$$

- C9. *Period-five internal branch $q \diamond x = q$: survives but constrained.* The right-tail fiber $\{a, q\} \rightarrow q$ must satisfy

$$(a \diamond q) \diamond a = (q \diamond q) \diamond q = q \setminus x, \quad a \diamond q \neq q \diamond q,$$

and the orbit recurrence also gives $b \diamond q = a$.

C10. *Period-five internal branch $q \diamond x = a$: survives but constrained.* The fiber $\{x, q\} \rightarrow a$ forces

$$(q \diamond a) \diamond q = b \diamond x = a \setminus x, \quad q \diamond a \neq b,$$

and the orbit recurrence gives $b \diamond a = a$.

C11. *Period-five branch $q \diamond x = p$: survives only as a split family.* It must satisfy

$$b \diamond p = a, \quad q \diamond (p \diamond x) = b, \quad p \diamond x \notin \{x, q\}.$$

If no external value appears at $p \diamond x$, then

$$p \diamond x \in \{a, b, p\},$$

with the corresponding extra tests listed in Proposition 19.

C12. *Period-six orbit-only branch: survives with collision.* For $O_6 = \{x, a, b, c, q, p\}$, a principal-orbit-only right tail maps the six elements of O_6 into $O_6 \setminus \{x\}$. Hence $R_x|_{O_6}$ has a collision, and every such collision must satisfy Proposition 15. The bad-edge exclusions from Proposition 13 require

$$b \diamond x \notin \{x, a\}, \quad c \diamond x \notin \{x, b\}, \quad q \diamond x \notin \{x, c\}, \quad p \diamond x \notin \{x, q\}.$$

Remark 21 (Stress-test ledger). The most compact surviving internal targets are the period-five branches $q \diamond x = q$ and $q \diamond x = a$. Both create a collision fiber inside the principal orbit and therefore expose explicit splitter equations. The period-four survivors are less compact because they force an external element, so the next hand-pruning pass should first apply Lemma 7 to the two period-five splitter rectangles, then return to the period-four external elements r and s .

6 Bad-point checklist and open gates

Proposition 22 (Bad-point package). *If E255 at x fails, then:*

1. *the L_x -orbit of x has period $d \geq 4$;*
2. *no u satisfies $u \diamond x = x$, so $x \notin \text{im } R_x$;*
3. *R_x has a canonical nontrivial collision fiber from Lemma 16;*
4. *on a fiber $R_x^{-1}(e)$ as above, $t \mapsto t \diamond e$ is injective and obeys (R) ;*
5. *H_x and F_x are fixed-point-free;*
6. *if $d = 4$, some one of $q \diamond x, p \diamond x$ is outside $\{x, a, q, p\}$.*

Proof. Proof certificate. Items: Corollary 9; Proposition 4; Lemma 16; Proposition 15; Lemma 6; Proposition 12. □

Remark 23 (Dependency table for a bad point). The following implications are the current compressed route graph. They are all proved above.

$$\begin{aligned}
\text{E255 at } x &\iff q \diamond x = x \\
&\iff \exists u (u \diamond x = x) \\
&\iff q \setminus x = x \\
&\iff ((x \diamond x) \diamond q) \diamond (x \diamond x) = x \\
&\iff (q \diamond x) \diamond q = p.
\end{aligned}$$

Under the negation $q \diamond x \neq x$:

$$\begin{aligned}
\text{no } u \diamond x = x &\implies x \notin \text{im } R_x \\
&\implies R_x \text{ has a nontrivial fiber} \\
&\implies y \diamond x = z \diamond x = e, \ y \neq z \\
&\implies (y \diamond e) \diamond y = (z \diamond e) \diamond z = e \setminus x, \quad y \diamond e \neq z \diamond e.
\end{aligned}$$

The canonical way to choose the fiber is the first repetition in the right tail $\rho_{n+1} = \rho_n \diamond x$. This removes choice from the splitter route.

For the orbit route:

$$\begin{aligned}
q \diamond x \neq x &\implies d \geq 4, \\
d = 4 &\implies q \diamond x \in \{p\} \cup (M \setminus \{x, a, q, p\}), \\
d = 4, \ q \diamond x = p &\implies p \diamond x \notin \{x, a, q, p\}.
\end{aligned}$$

So a bad period-four point exits the principal orbit along the right tail by the step $p \diamond x$. No four-element principal-orbit-only bad point survives these facts.

Remark 24 (Finite facts used, nothing else). Only these finite principles appear: finite surjection = bijection; a finite self-map missing a point has a nontrivial fiber; every finite forward orbit repeats; and conjugate finite self-maps have the same cycle structure. Every cancellation in the paper is left cancellation by a proved bijective left translation.

Remark 25 (Do not use).

No right cancellation. No associativity. No identity element.
No universal idempotence. No global “ H_x has no cycles”.
Do not use $d_k \setminus x = c_{k-2} \diamond c_{k-1}$.
Do not use $p \diamond d_k = d_{k+1}$ unless freshly proved.

Gate 26 (Fixed-point gate). Prove $q \diamond x = x$, equivalently Lemma L or $(q \diamond x) \diamond q = p$, with a visible finite step.

Gate 27 (Splitter gate). Use the canonical right-tail collision fiber, or any collision $d_i = d_j = e$, to turn the injective splitter $t \mapsto t \diamond e$ into either $q \diamond x = x$ or a finite counting contradiction.

Gate 28 (Period-four external gate). At a bad period-four point, feed the forced external element among

$$q \diamond x, \quad p \diamond x$$

back into the fixed-point gate or the splitter gate.

Gate 29 (Period-five pruning gate). Use the explicit internal splitter rectangles in Proposition 19, especially the branches $q \diamond x = q$ and $q \diamond x = a$, to force either $q \diamond x = x$, an external product, or a finite counting contradiction.

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